Environ Monit Assess
DOI 10.1007/s10661-009-1265-2

Seasonal and spatial variation of Yamuna River water quality in Delhi, India

Papiya Mandal · Rahul Upadhyay · Aziz Hasan

Received: 4 August 2009 / Accepted: 3 December 2009
© Springer Science+Business Media B.V. 2009

Abstract Yamuna river pollution has been extensively studied with regard to some selected parameters in five locations at Palla, Nizamuddin Midstream, Nizamuddin Quarter Stream, Agra Canal Midstream, and Agra Canal Quarter Stream in Delhi, India. Seasonal and location-wise variation of pollutants namely dissolve oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), total Kjeldahl nitrogen (TKN), ammonia (AMM), total coliform and fecal coliform were studied for 6 years during the period of 2000–2005. The study results revealed the lowest level of pollution during monsoon. The statistical analysis revealed a positive correlation between DO, BOD, COD, TKN, and AMM.

Keywords Spatial variation · Water quality · Yamuna River · Pollution · Correlations · Regression

P. Mandal (✉)
Delhi Zonal Laboratory,
Council of Scientific and Industrial Research (CSIR),
National Environmental Engineering Research Institute, CSIR Building, A-93/94, Phase I,
Naraina Industrial Area, New Delhi 110 028, India
e-mail: papiya.mandal 1942@gmail.com,
p_mandal@neeri.res.in, papiya.mandal@yahoo.com

R. Upadhyay
S S L Jain PG Collage,
Vidisha, Madhya Pradesh, India
e-mail: rahul_earthlover@yahoo.co.in

A. Hasan
Analytical Instruments Division, NEERI, CSIR,
Nehru Marg, Nagpur 440 020, India
e-mail: azizhasan@live.in

Introduction

It is well known that clean water is absolutely essential for several purposes for healthy living. The river Yamuna is subjected to multiple uses for community water supply, irrigation, industrial water supply, bathing, and disposal of sewage and industrial effluents. It is an only natural resource for sustaining all forms of life in Delhi, but perennial increase of population and urban activities in Delhi are placing tremendous pressures and demands on this natural resource. There is a heavy pressure of water supply and sanitation on river Yamuna in Delhi. The chemistry of river waters is dictated by supply of various elements from both natural and anthropogenic sources (Krishnaswami and Singh 2005). Rainfall plays an active role for changing the water quality of the river Yamuna. The climate of the Delhi region is semiarid type with three defined seasons, e.g., winter, summer, and monsoon. The annual average temperature recorded in Delhi is usually between 31.5°C based on records over the period

Published online: 29 December 2009

of 70 years maintained by the Meteorological Department. Delhi receives a total rainfall around 611.8 mm/year of which 87% of annual rainfall is received during monsoon month June to September (CSE 2008). Monthly rainfall intensity of Delhi city is shown in Fig. 1. River water quality is the composite of several interrelated parameters which are subjected to local and temporal variations and also affected by volume of water flow and discharge rates. River water quality study is usually required for stabilizing base line conditions, setting quality criteria, and standards (Ahmed 2004). Yamuna river pollution in Delhi stretch is gaining interest to Delhi habitants because of their public health impact and other issues. In order to study the effect of wastewater disposal on river Yamuna, continuous monitoring and assessment are required. The Central Pollution Control Board (CPCB) has carried out routine monitoring of Yamuna water quality study on monthly basis during the period from 2000 to 2005. The paper will meet the following objectives.

- To assess the monthly/seasonal variation of selected parameters like dissolve oxygen (DO), chemical oxygen demand (COD), biochemical oxygen demand (BOD), total Kjeldahl nitrogen (TKN), ammonia (AMM), total coliform (TC), and fecal coliform (FC) at five selected locations namely Palla, Nizamuddin Mid Stream (NMS), Nizamuddin Quarter Stream (NQS), Agra Canal Midstream (ACMS), Agra Canal Quarter Stream (ACQS)
- To prepare the comparative statements of interrelationship of the selected parameters
- To compare the results with standards

In this paper, an attempt has been taken to study the monthly/seasonal and spatial variation of water quality of river Yamuna. The sampling location has been identified in Fig. 2.

Materials and methods

The various water quality analyses were carried out following the 20th edition of the Standard Methods for Examination of Water and Wastewater, published by APHA, AWWA, WEF (1998). Parameter-wise measurement methods are describes in Table 1. The site descriptions are as given below.

Pearson's correlation and regression were used to detect linear correlations and regressions between selected parameters. Table 2 shows summarized location-wise correlation and regression of selected parameters.

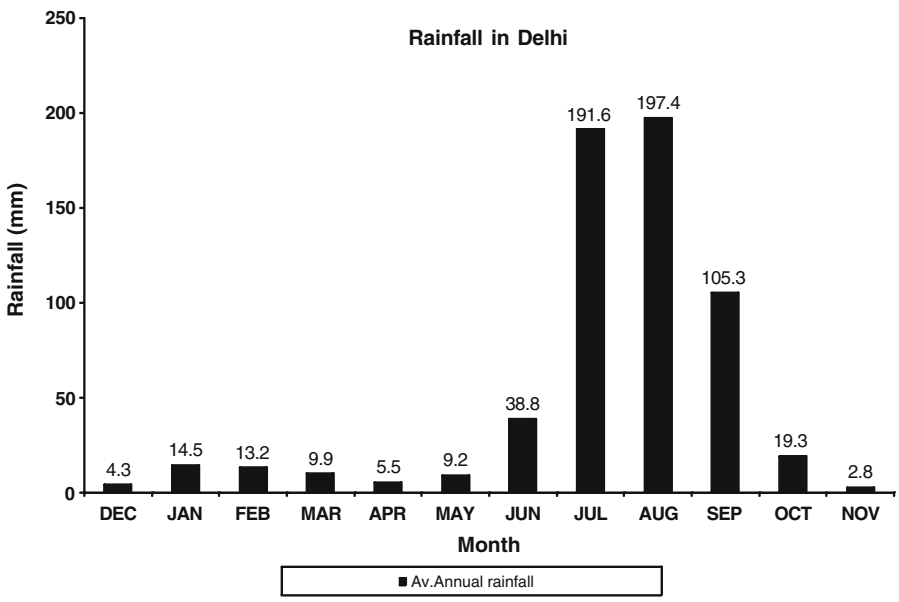

Fig. 1 Monthly rainfall intensity of Delhi City

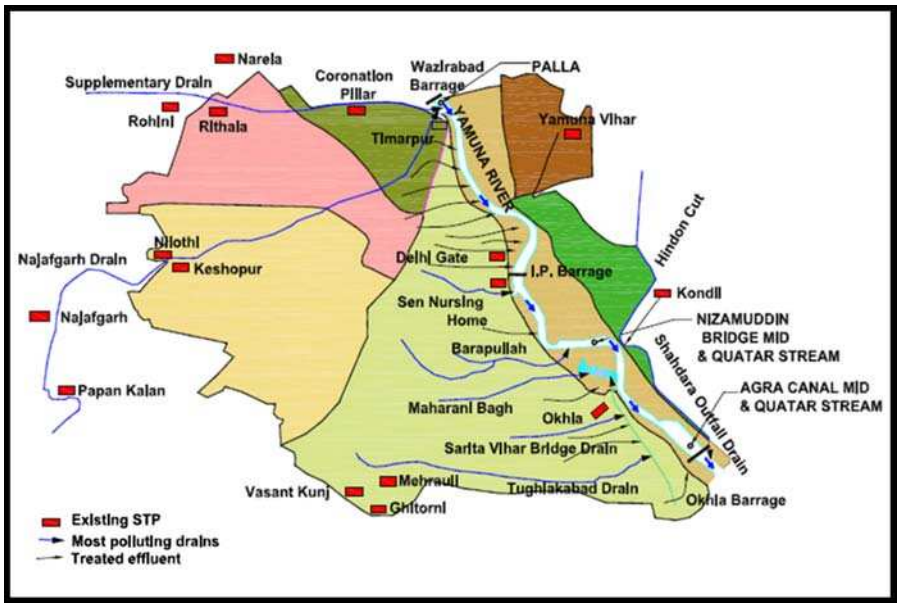

Fig. 2 Sampling location of River Yamuna at Delhi stretch

Site descriptions

The monitoring sites were selected by CPCB on the basis of need and potential of water quality import or pollution load transported, respectively.

- *Palla*: About 15 km upstream from Wazirabad barrage near cremation ground. Sampling at this location reflects the water quality before receiving the wastewater discharges from Delhi and raw water quality for Delhi's water supply. Water quality at this location also reflects the impact of domestic and industrial discharges from Sonepat District of Haryana.
- *Nizamuddin Bridge Midstream and Nizamuddin Bridge Quarter Stream*: 13 km downstream from Wazirabad barrage (near Sarai Kale Khan bus stand) at Delhi—Ghaziabad (Noida) Road Bridge on Yamuna River. The water quality at this location reflects the impact of wastewater discharge. Nizamuddin Bridge Midstream and Nizamuddin Bridge Quarter Stream were within few meters. These two sampling locations have been chosen for accuracy of pollution load at selected stretch.
- *Agra Canal Midstream and Agra Canal Quarter Stream*: 24 km downstream from Wazirabad barrage and East of Delhi–Agra national highway near Madanpur Khaddar village. The water quality at this location reflects the impact of discharge of treated and partially treated effluents from Okhla Sewage treatment Plant. Agra Canal Midstream and Agra Canal Quarter Stream were within few meters. These two sampling locations have

Table 1 Parameter-wise measurement methods

Parameters	Unit	Measurement methods
Dissolved oxygen (DO)	mg/L	Winkler modified method
Biochemical oxygen demand (BOD)	mg/L	DO consumption in 3 days at 27°C
Chemical oxygen demand (COD)	mg/L	Potassium dichromate method
Ammonia	mg/L	Nesslerization method
Kjeldahl nitrogen	mg/L	Kjeldahl method
Total coliform (TC)	No./100 mL	MF technique
Fecal coliform (FC)	No/100 mL	MF technique

Table 2 Correlation coefficient and regression of selected parameters at various points of Yamuna River in Delhi

Correlation coefficient and regression	Palla	NMS	NQS	ACMS	ACQS
r(BOD,COD)	0.06	0.93	0.98	0.96	0.95
Regression	COD = 0.2271 BOD + 10.413	COD = 2.3255 BOD + 13.32	COD = 2.5074 BOD + 9.816	COD = 3.274 BOD + 5.044	COD = 3.1639 BOD + 6.79
	$R^2 = 0.004$	$R^2 = 0.873$	$R^2 = 0.960$	$R^2 = 0.920$	$R^2 = 0.909$
r(BOD,DO)	0.00	(−)0.90	(−)0.90	(−)0.80	(−)0.80
Regression	DO = 0.0504 BOD + 7.945	DO = −0.111 BOD + 3.152	DO = −0.1219 BOD + 3.537	DO = −0.1582 BOD + 3.028	DO = −0.2057 BOD + 3.74
	$R^2 = 0.003$	$R^2 = 0.849$	$R^2 = 0.795$	$R^2 = 0.588$	$R^2 = 0.720$
r(BOD,TKN)	(−)0.47	0.83	0.89	0.81	0.89
Regression	TKN = 0.2437 BOD + 2.388	TKN = 0.7247 BOD + 3.970	TKN = 0.8236 BOD + 2.220	TKN = 0.8749 BOD + 2.618	TKN = 0.9763 BOD + 1.79
	$R^2 = 0.221$	$R^2 = 0.684$	$R^2 = 0.786$	$R^2 = 0.654$	$R^2 = 0.788$
r(BOD,AMM)	0.09	0.86	0.88	0.83	0.94
Regression	AMM = 0.0126 BOD + 0.414	AMM = 0.7006 BOD − 0.625	AMM = 0.7549 BOD − 1.473	AMM = 0.7612 BOD − 0.429	AMM = 0.7868 BOD − 0.84
	$R^2 = 0.008$	$R^2 = 0.745$	$R^2 = 0.769$	$R^2 = 0.687$	$R^2 = 0.877$
r(COD,DO)	(−)0.70	(−)0.90	(−)0.92	(−)0.79	(−)0.82
Regression	DO = −0.1923 COD + 10.22	DO = −0.0435 COD + 3.518	DO = −0.0489 COD + 4.034	DO = −0.0476 COD + 3.238	DO = −0.0601 COD + 3.94
	$R^2 = 0.484$	$R^2 = 0.801$	$R^2 = 0.838$	$R^2 = 0.620$	$R^2 = 0.678$
r(COD,TKN)	0.10	0.85	0.93	0.88	0.86
Regression	TKN = 0.0149 COD + 1.764	TKN = 0.3007 COD + 0.512	TKN = 0.3371 COD − 1.572	TKN = 0.2778 COD + 0.757	TKN = 0.2856 COD + 0.843
	$R^2 = 0.011$	$R^2 = 0.730$	$R^2 = 0.863$	$R^2 = 0.768$	$R^2 = 0.743$
r(COD,AMM)	0.01	0.86	0.93	0.88	0.92
Regression	COD = 0.3692 AMM +8.444	AMM = 0.2819 COD +3.411	AMM = 0.3142 COD −5.294	AMM = 0.2364 COD −1.794	AMM = 0.233 COD −1.75
	$R^2 = 0.188$	$R^2 = 0.748$	$R^2 = 0.873$	$R^2 = 0.772$	$R^2 = 0.846$
r(DO,TKN)	(−)0.27	(−)0.74	(−)0.84	(−)0.67	(−)0.67
Regression	TKN = −0.1402 DO + 3.0524	TKN = −5.396 DO + 23.686	TKN = −5.6739 DO + 25.23	TKN = −3.5146 DO + 17.526	TKN = −3.0419 DO + 17.92
	$R^2 = 0.075$	$R^2 = 0.550$	$R^2 = 0.698$	$R^2 = 0.40$	$R^2 = 0.450$
r(DO,AMM)	(−)0.29	(−)0.81	(−)0.88	(−)0.69	(−)0.77
Regression	AMM = −0.0412 DO + 0.769	AMM = −5.4832 DO + 18.64	AMM = −5.5151 DO + 19.875	AMM = −3.0873 DO + 12.57	AMM = −2.6678 DO + 12.36
	$R^2 = 0.086$	$R^2 = 0.663$	$R^2 = 0.768$	$R^2 = 0.481$	$R^2 = 0.592$
r(TKN,AMM)	0.02	0.96	0.96	0.95	0.90
Regression	AMM = 0.0042 TKN + 0.421	AMM = 0.8869 TKN − 2.899	AMM = 0.8877 TKN − 2.913	AMM = −0.8025 TKN − 1.74	AMM = −0.6862 TKN − 0.484
	$R^2 = 0.0002$	$R^2 = 0.917$	$R^2 = 0.917$	$R^2 = 0.894$	$R^2 = 0.806$

been chosen for accuracy of pollution load at selected stretch.

Results and discussion

Monthly/seasonal variations of selected parameters during the period of 2000 to 2005 are shown in Figs. 3, 4, 5, 6, 7, 8, and 9. Standard deviation (SD) of the 6-year data reflects that the SD was less for DO, BOD, COD, TKN, and AMM and was high for TC and FC. SD of selected parameters was not incorporated in the paper.

DO drops to alarming levels at middle and downstream of river Yamuna shown in Fig. 3, the attainment of maximum level of DO during monsoon (June to August). The maximum and minimum concentrations of DO at Palla were 9.1 and 6.6 mg/L. DO content, which plays a vital role in supporting aquatic life in running water, is susceptible to slight environmental changes (Ahmed 2004). Very low DO may result in anaerobic conditions that cause bad odors.

BOD is an indicator of organic pollution. Unpolluted natural waters have a BOD of 5 mg/L or less (Schulze et al. 2001). The highest average BOD level measured at Nizamuddin stretch is shown in Fig. 4. The maximum permissible level of BOD for bathing is 3 mg/L. BOD was inversely related to DO.

Spatial variation of COD is shown in Fig. 5. COD is positively correlated with BOD, TKN, and AMM and inversely related to DO. High COD interferes with oxygen transfer to the soil, thus affecting plant growth (Adekunle et al. 2007). Drinking water supply should not exceed COD of 2.5 mg/L, and potable water of COD content greater than 7.5 mg/L is regarded as poor (Esa 1983).

Spatial variations of TKN and AMM are shown in Figs. 6 and 7. Ammonia in the form of nitrogen is readily available to marine life and microorganisms. The presence of ammonia and nitrogenous sources in large amounts may result in eutrophication, leading to algal blooms and anoxic conditions (Elmanama et al. 2006). TKN and AMM were highly correlated. Correlation at Nizamuddin Stream and Agra Canal Stream of river Yamuna was more than 0.8. Presence of nutrients under normal conditions supports the growth of bacteria and other microorganisms, leading to higher BOD levels as well as a correspond-

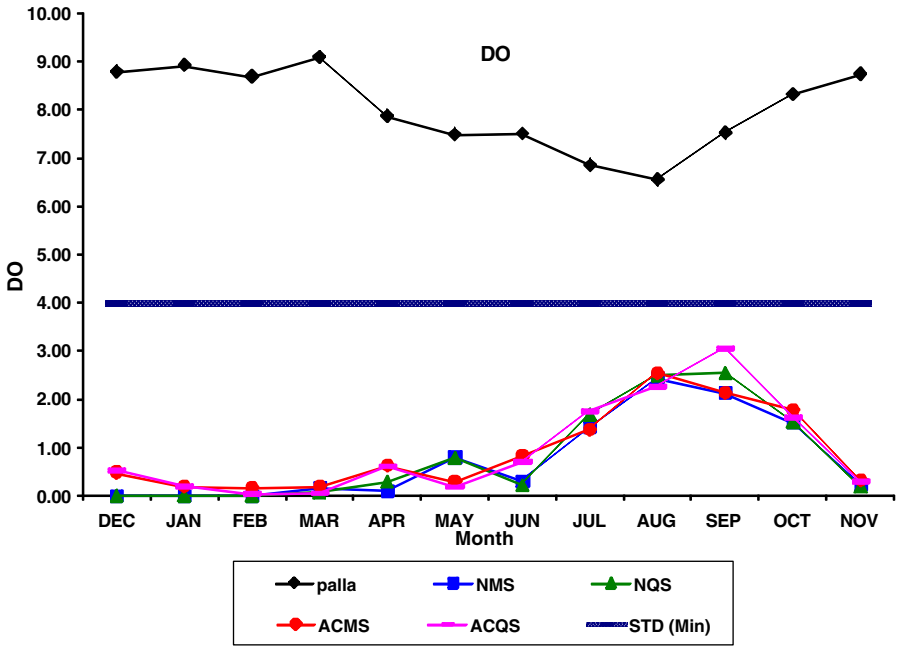

Fig. 3 Spatial variation of DO of Yamuna River at different location of Delhi Stretch during the sampling period

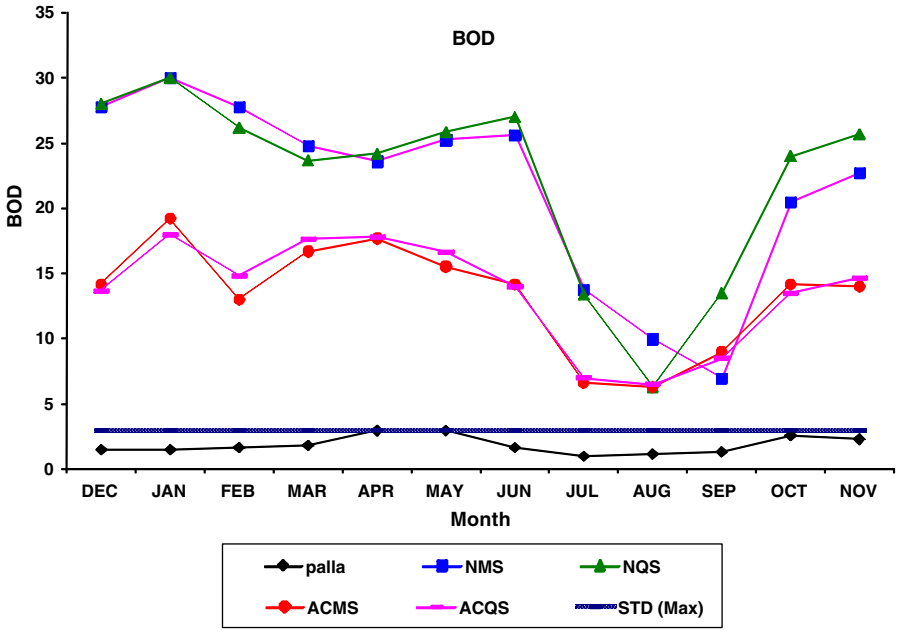

Fig. 4 Spatial variation of BOD of Yamuna River at different location of Delhi Stretch during the sampling period

ing increase in the fecal coliform count. This finding was supported by a different researcher (Jannasch 1968).

TC and FC are indicators for pathogenic organisms. They are usually present in surface water, soil, and feces of humans and animals. Monthly/seasonal and spatial variations of TC and FC are shown in Figs. 8 and 9. It has been noticed that, even at Palla, upstream of river Yamuna, desired criteria of water quality regarding TC were above the permissible limit of CPCB Standard (Water Quality Criteria, CPCB 2009).

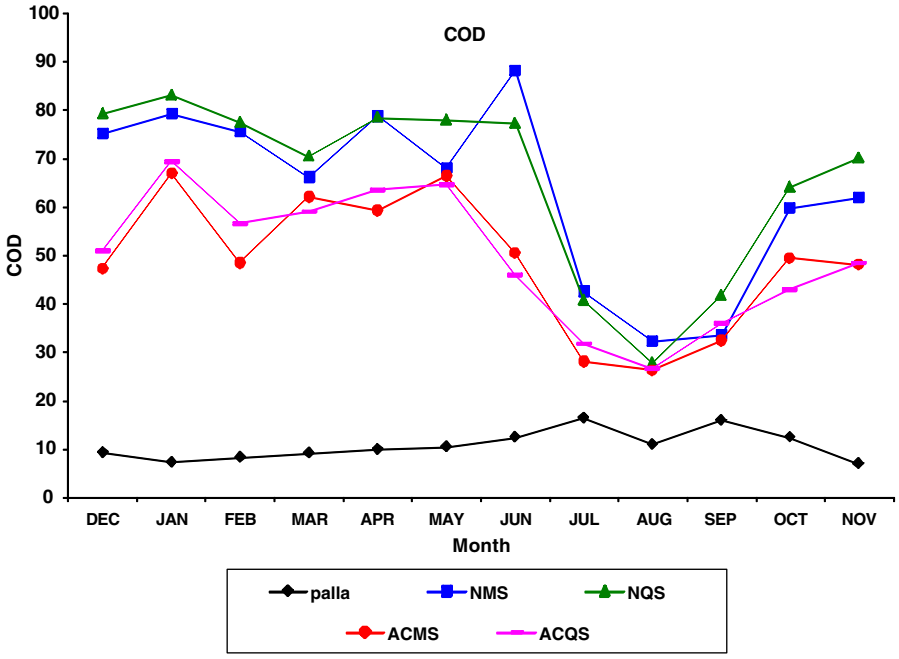

Fig. 5 Spatial variation of COD of Yamuna River at different location of Delhi Stretch during the sampling period

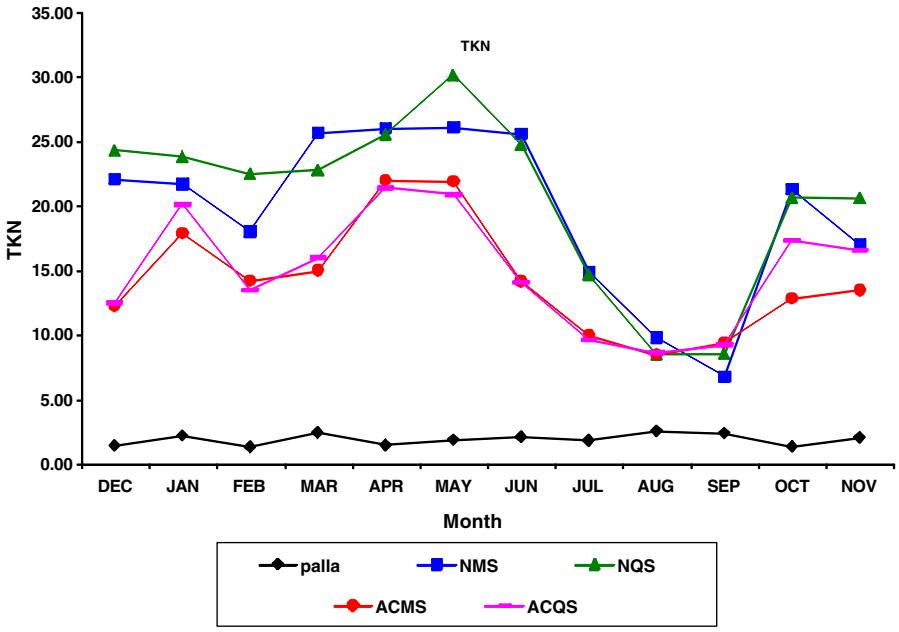

Fig. 6 Spatial variation of TKN of Yamuna River at different location of Delhi Stretch during the sampling period

Spatial variation

The results indicated no significant difference in the measured concentration of pollutants at all locations during winter and summer months, October to May. Significant differences were noted during monsoon months, June to September. The flow of the Yamuna River varies significantly during monsoon and nonmonsoon seasons. DO, BOD, COD, TKN, and AMM were varied significantly with locations and seasons. TC and FC concentrations were significantly high at Nizamuddin and Agra Canal Stretch during the summer season in the month of May. The total and fecal

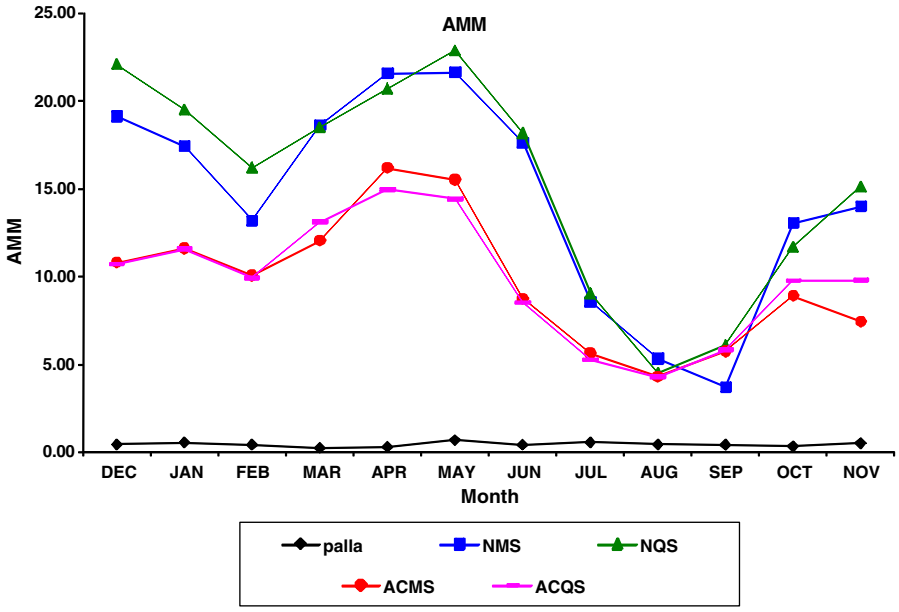

Fig. 7 Spatial variation of AMM of Yamuna River at different location of Delhi Stretch during the sampling period

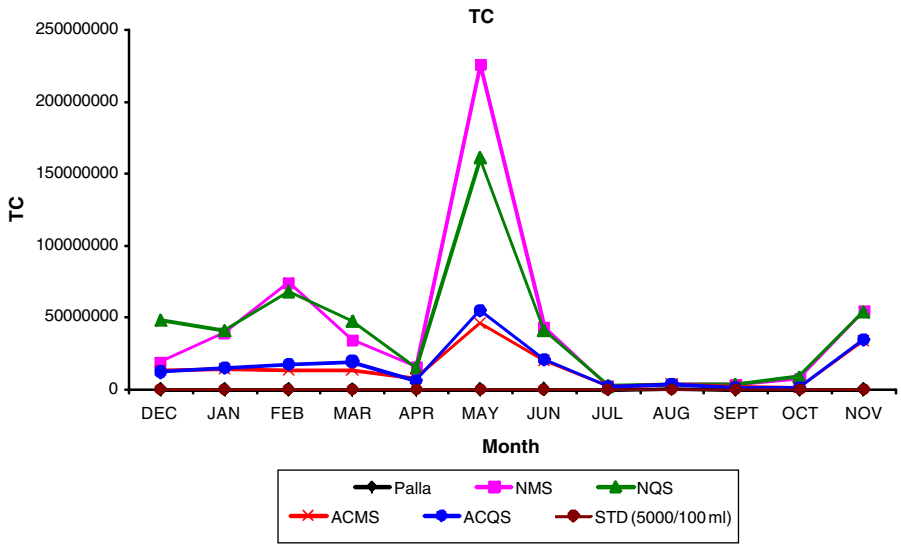

Fig. 8 Spatial variation of TC of Yamuna River at different location of Delhi Stretch during the sampling period

coliform reduced significantly during monsoon but was much higher than water quality standard (Water Quality Status of Yamuna River, CPCB 2006).

Overall, Palla was meeting all desired criteria for selected water quality parameters. Palla had appreciable DO and low BOD, COD, TKN, AMM, TC, and FC as compared to other selected sites. Water at Palla could be classified as "C" category according to CPCB standard shown in Table 3. Water at Nizamuddin and Agra Canal Stretch cannot be use even for bathing purposes also. A 22-km stretch of Yamuna River in Delhi is not only dead but an overload of coliform contamination (CSE 2009). Under the Yamuna Action plan, Phase I, 1993, a total of 16 upflow Anaerobic Sludge Blanket Reactors (UASBR) were constructed along river Yamuna. The cost of construction, operation and maintenance cost, and low energy consumptions were the addi-

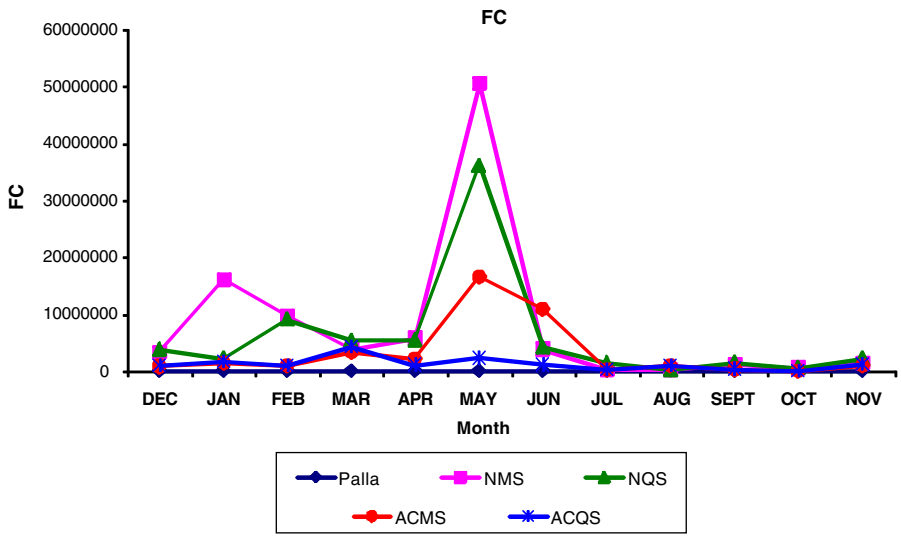

Fig. 9 Spatial variation of FC of Yamuna River at different location of Delhi Stretch during the sampling period

Table 3 Designated best use classification of surface water

Designated best use	Quality class	Primary water quality criteria
Drinking water source without conventional treatment but with chlorination	A	Total coliform organisms (MPN[a]/100 ml) shall be 50 or less pH between 6.5 and 8.5 Dissolved oxygen 6 mg/L or more Biochemical oxygen demand 2 mg/L or less
Outdoor bathing (organized)	B	Total coliform organisms (MPN/100 ml) shall be 500 or less pH between 6.5 and 8.5 Dissolved oxygen 5 mg/L or more Biochemical oxygen demand 3 mg/L or less
Drinking water source with conventional treatment	C	Total coliform organisms (MPN/100 ml) shall be 5,000 or less pH between 6 and 9 Dissolved oxygen 4 mg/L or more Biochemical oxygen demand 3 mg/L or less
Propagation of wildlife and fisheries	D	pH between 6.5 and 8.5 Dissolved oxygen 4 mg/L or more Free ammonia (as N) 1.2 mg/L or less
Irrigation, industrial cooling, and controlled disposal	E	pH between 6.0 and 8.5 Electrical conductivity less than 2,250 μmho/cm Sodium absorption ratio less than 26

Source: Guidelines for water quality monitoring, MINARS/2007-08 (CPCB 2007)
MPN most probable number

tional main advantage, but the major problem with UASBR is that these plants produced very poor quality of effluent having high level of BOD, COD, and FC (Mittal 2008). Poor condition of trunk sewers, shortage of sewage treatment capacity, and lack of sanitation facilities in unsewered area of Delhi which account for nearly 50% of population are responsible for continued pollution of Yamuna in Delhi. Delhi alone contributes around 3,296 MLD/day of sewage by virtue of drains out falling in Yamuna Delhi segment flow (22-km length). This is more than that of all the class 2 cities of India put together (Jain et al. 2007). The river Yamuna receives discharge from 22 drains at downstream of Wazirabad. The water quality monitoring of 22 drains in Delhi indicates that except Khyber Pass drain 21 drains were not meeting the criteria for one of the other indicator parameters, i.e., COD, BOD, and suspended solids (Bhardwaj et al. 2003). Blockage of drains by solid waste reduces the carrying capacity of drains and natural canals and also a source of pollution (Alam et al. 2007). The Najafgarh drain falls at Yamuna before Nizamuddin stretch which is a major contributor to the pollution of river (Bhardwaj et al. 2003). So Nizamuddin stretch might be the most polluted among all three selected sites in Delhi.

Conclusions

The results of this study demonstrated that the untreated and partially treated wastewater disposed to river Yamuna is highly polluted. Various environmental parameters are varied with location and season. Recycle and reuse of wastewater as water after providing minimum treatment can reduce the wastewater discharge into river Yamuna. Providing artificial DO to the polluting drains will reduce significantly BOD and other associated pollution load to the river. Identification of unsewered areas and provide an effective sewerage system may result in treated sewage, out falling

in the river through open drains/nallahs. Efficient and sophisticated treatment plant for domestic and industrial sewage to ensure their conformity with the standards prescribed by Delhi Pollution Control Committee under the provisions of the Water (Prevention and Control of Pollution) Act, 1974 will significantly reduce the pollution load of Yamuna. Avoiding mixing of treated and untreated domestic as well as industrial sewage into surface drainage may reduce the pollution load to river. Understanding the seasonal variation and interrelationship of the selected parameters may be helpful in routine analysis of river water quality.

Acknowledgements The authors are grateful to the Central Pollution Control Board (CPCB), New Delhi, India for providing the relevant data of Yamuna River water quality. Authors are also grateful to the Acting Director Dr. T. Chakrabarti, National Environmental Engineering Research Institute (NEERI), for granting permission to publish this work.

References

Adekunle, I. M., Adetunji, M. T., Gbadebo, A. M., & Banjoko, B. O. (2007). Assessment of groundwater quality in a typical rural settlement in Southwest Nigeria. *International Journal of Environmental Research and Public Health, 4*(4), 307–318.

Ahmed, A. (2004). Ecological studies of the River Padma at Mawa Ghat, Munshiganj I. Physicochemical properties. *Pakistan Journal of Biological Sciences, 7*(11), 1865–1869.

Alam, J. B., Hossain, A., Khan, S. K., Banik, B. K., Islam, M. R., Muyen, Z., et al. (2007). Deterioration of water quality of Surma river. *Environmental Monitoring and Assessment, 134*, 233–242.

APHA (1998). *American public health association. Standard methods for examination of waters and wastewaters* (20th ed.). Washington, DC: APHA.

Bhardwaj, R. M., Mehrotra, K., Trivedi. R. C., Makhijani, S. D. (2003). Status of water quality in Yamuna River. In *Paper presented at the seminar on water for national capital territory, Delhi*.

CPCB (2006). Water quality status of Yamuna River. Assessment and development study of River Basin series: ADSORBS/41/2006-07. http://www.cpcb.nic.in/newitems/11.pdf. Accessed 11 January 2009.

CPCB (2007). Guidelines for water quality monitoring, MINARS/27/2007-08. http://www.cpcb.nic.in/upload/NewItems/NewItem_116_Guidelinesof%20waterqualitymonitoring_31.07.08.pdf. Accessed on 4 November 2009

CPCB (2009). Water quality criteria. http://cpcbenvis.nic.in/waterpollution/criteria.htm. Accessed on 4 November

CSE (2008). Climate & rainfall. http://www.rainwaterharvesting.org/index_files/climate_rainfall.htm. Accessed 10 January 2009.

CSE (2009). State of pollution in the Yamuna. http://www.indiaenvironmentportal.org.in/files/State%20of%20the%20Yamuna_0.pdf. Accessed on 16 July 2008.

Elmanama, A. A., Afifi, S., & Bahr, S. (2006). Seasonal and spatial variation in the monitoring parameters of Gaza Beach during 2002–2003. *Environmental Research, 101*(1), 25–33.

Esa, R. (1983). *Drilled wells and groundwater in the Precambrian crystalline bedrock of Finland* (No. 52, p. 57). Helsinki: Water Research Institute.

Jain, S. K., Agarwal, P. K., & Singh V. P. (2007). *Hydrology and water resources of India. Water science and technology library* (Vol. 57). Dordrecht: Springer.

Jannasch, H. (1968). Competitive elimination of Enterobacteriaceae from sea water. *Applied Microbiology, 16*, 1616–1618.

Krishnaswami, S., & Singh, S. K. (2005). Chemical weathering in the river basins of the Himalaya, India. *Current Science (Special Section: Water), 89*(5), 841–849.

Mittal, A. K. (2008). Urban run-off: Source of major microbial load on Indian Rivers. (December 15–16) paper presented at the national Conference on Hydraulics, *Water Resources and Environment, HYDRO-2008*. Jaipur City

Schulze, D. L., Eggers, J. S., Esser, L. R. (2001). Water quality studies red rock and Saylorville Reservoirs Des Moines River, Lowa. Annual Report. Department of the Army, Rock Island District, Corps of Engineers, Rock Island, IL.

View publication stats